

SIMPLE LINEAR REGRESSION

Complete Assignment Answers

Question 1: What is Simple Linear Regression (SLR)? Explain Its Purpose.

Simple Linear Regression (SLR) is a statistical and Machine Learning technique used to study and model the relationship between **one independent variable (X)** and **one dependent variable (Y)**.

It fits a straight line to the data and uses that line to estimate or predict the value of Y based on X.

Purpose of SLR

The main purposes are:

1. To understand the relationship between two variables.
2. To predict the dependent variable.
3. To measure how changes in X are associated with changes in Y.
4. To estimate the strength and direction of the relationship.

Example

If we want to predict **salary based on years of experience**:

Years of Experience → Salary

Years of experience is X, and salary is Y.

Easy Definition

Simple Linear Regression predicts one dependent variable using one independent variable through a straight-line relationship.

Question 2: What Are the Key Assumptions of Simple Linear Regression?

The major assumptions are:

1. Linearity

There should be an approximately linear relationship between X and Y.

2. Independence

The observations and regression errors should be independent of each other.

3. Homoscedasticity

The variance of the residuals should remain approximately constant across the range of fitted values.

If the variance changes, it is called **heteroscedasticity**.

4. Normality of Residuals

For conventional statistical inference, the residuals are generally assumed to be approximately normally distributed.

5. No Problematic Outliers

Extreme or influential observations should not disproportionately affect the regression line.

Summary

Assumption	Meaning
Linearity	X and Y have a linear relationship
Independence	Observations/errors are independent
Homoscedasticity	Constant residual variance
Normality	Residuals are approximately normal
No influential outliers	Extreme points should not dominate the model

Easy Definition

SLR assumes linearity, independence, constant error variance, approximately normal residuals for inference, and no problematic influential observations.

Question 3: Write the Mathematical Equation for a Simple Linear Regression Model and Explain Each Term.

The mathematical equation is:

$$Y = \beta_0$$

$$+ \beta_1 X + \epsilon$$

For prediction, we commonly write:

$$\hat{Y} = \beta_0 + \beta_1 X$$

Explanation of Terms

Term	Meaning
(Y)	Actual dependent variable
$(\hat{Y})$	Predicted value of Y
(X)	Independent variable
(β_0)	Intercept
(β_1)	Slope/coefficient
(ϵ_n)	Error term

Intercept

$$\beta_0$$

is the expected value of Y when X is 0.

Slope

$$\beta_1$$

represents the expected change in Y for a **one-unit increase in X**.

Example

Suppose:

$$\hat{Y} = 10 + 5X$$

Here:

- Intercept = 10
- Slope = 5

If X increases by 1, predicted Y increases by 5.

Question 4: Provide a Real-World Example Where Simple Linear Regression Can Be Applied.

Example: House Price Prediction

A real-estate company can use Simple Linear Regression to predict the price of a house based on its **area**.

Independent Variable (X) → House Area

Dependent Variable (Y) → House Price

Suppose the model is:

$$\text{Price} = 50000 + 2000(\text{Area})$$

The model can estimate the price for a house based on its area.

Other Examples

- Study hours → Exam marks
- Years of experience → Salary
- Advertising expenditure → Sales
- Temperature → Electricity demand
- Distance travelled → Fuel consumption

Easy Example to Remember

Study Hours → Exam Marks

Question 5: What Is the Method of Least Squares in Linear Regression?

The **Least Squares Method** is a technique used to find the best-fitting regression line by minimizing the **sum of squared residuals**.

A residual is:

$$e_i = y_i - \hat{y}_i$$

The least-squares method minimizes:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

In simple linear regression:

$$\hat{y}_i =$$

$$\beta_0 + \beta_1 x_i$$

Therefore, the method chooses the intercept and slope that make the total squared prediction error as small as possible.

Why Square the Errors?

Squaring:

- Makes all errors positive.
- Gives greater weight to larger errors.
- Provides a convenient mathematical optimization objective.

Example

If actual value = 100 and predicted value = 90:

$$\text{Error} = 100 - 90 = 10$$

Squared error:

$$10^2 = 100$$

The regression algorithm tries to choose the line that minimizes the total squared error across all observations.

Easy Definition

Least Squares = Find the regression line that minimizes the sum of squared prediction errors.

Question 6: What Is Logistic Regression? How Does It Differ from Linear Regression?

Logistic Regression is a supervised Machine Learning algorithm primarily used for **classification**, especially binary classification.

It predicts the probability of an observation belonging to a class.

For binary classification, the logistic/sigmoid function maps a linear combination of inputs to a value between 0 and 1:

$$p = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 X$$

Example

Predict whether a customer will purchase a product:

1 → Purchase

0 → No Purchase

Logistic vs Linear Regression

Linear Regression	Logistic Regression
Used mainly for regression	Used mainly for classification
Predicts continuous values	Predicts class probabilities
Output can be any real number	Probability is between 0 and 1
Uses a linear prediction equation	Uses a logistic/sigmoid transformation
Example: predict salary	Example: predict yes/no

Example

Linear Regression:

Predict house price = ₹5,00,000

Logistic Regression:

Probability customer buys = 0.82

A classification threshold can then be applied, such as 0.5, to convert probability into a class label.

Easy Definition

Linear Regression predicts continuous values, while Logistic Regression predicts probabilities/classes.

Question 7: Name and Briefly Describe Three Common Evaluation Metrics for Regression Models.

Three common regression evaluation metrics are:

1. Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values.

MAE=

1

n

$\sum$

i=1

n

|

y

i

-

y

^

i

|

Lower MAE indicates smaller average prediction errors.

2. Mean Squared Error (MSE)

MSE calculates the average squared prediction error.

MSE=

1

n

$\sum$

i=1

n

(

y

i

-

y

^

i

)

2

Because errors are squared, large errors receive more weight.

Lower MSE is better.

3. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE.

$$\text{RMSE} = \sqrt{\text{MSE}}$$

It is expressed in the same units as the target variable.

Lower RMSE indicates better predictive performance.

Comparison

Metric	Meaning	Lower is better?
MAE	Average absolute error	Yes
MSE	Average squared error	Yes
RMSE	Square root of MSE	Yes

Question 8: What Is the Purpose of the R-squared Metric in Regression Analysis?

R-squared (R^2) measures the proportion of variation in the dependent variable that is explained by the regression model.

A common formula is:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

Where:

- (SS_{res}) = residual sum of squares
- (SS_{tot}) = total sum of squares

Example

If:

$$R^2 = 0.80$$

then the model explains approximately **80% of the variation in the target variable within the evaluated dataset.**

Interpretation

- (R^2) closer to 1 → model explains more variation.
- (R^2) closer to 0 → model explains less variation.

For models with an intercept, (R^2) is commonly between 0 and 1, but it can be negative on test data or in other settings when the model performs worse than the baseline mean predictor.

Important Point

A high (R^2) does not automatically mean the model is good. We should also consider:

- MAE
- RMSE
- Residual plots
- Test-set performance
- Model assumptions
- Business context

Easy Definition

R^2 tells us how much of the variation in the target is explained by the regression model.

Question 9: Write Python Code to Fit a Simple Linear Regression Model Using Scikit-learn and Print the Slope and Intercept.

Python / Google Colab Code

```
import numpy as np

from sklearn.linear_model import LinearRegression
```

```
# Independent variable

X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)


# Dependent variable

Y = np.array([2, 4, 6, 8, 10])


# Create the model

model = LinearRegression()


# Fit the model

model.fit(X, Y)


# Print slope and intercept

print("Slope:", model.coef_[0])

print("Intercept:", model.intercept_)
```

Expected Output

Slope: 2.0

Intercept: 0.0

Therefore, the regression equation is:

$$\hat{Y} = 0 + 2X$$

or:

$$\hat{Y} = 2X$$

Explanation

```
model.fit(X, Y)
```

trains the linear regression model.

```
model.coef_[0]
```

returns the **slope**.

```
model.intercept_
```

returns the **intercept**.

Visualization

```
import matplotlib.pyplot as plt
```

```
Y_pred = model.predict(X)
```

```
plt.scatter(X, Y, label="Actual Data")
```

```
plt.plot(X, Y_pred, linewidth=2, label="Regression Line")
```

```
plt.xlabel("X")
```

```
plt.ylabel("Y")
```

```
plt.title("Simple Linear Regression")
```

```
plt.legend()
```

```
plt.show()
```

Question 10: How Do You Interpret the Coefficients in a Simple Linear Regression Model?

Consider the model:

$$\hat{Y} = \beta$$

$$\beta_0 + \beta_1 X$$

There are two main coefficients.

1. Intercept (β_0)

The intercept represents the predicted value of Y when X is equal to 0.

Example:

$$\hat{Y} = 50 + 10X$$

Here:

$$\beta_0 = 50$$

So when $X = 0$, predicted $Y = 50$.

However, the intercept is only substantively meaningful if $X = 0$ is within a sensible range of the data.

2. Slope (β_1)

The slope represents the expected change in predicted Y for a **one-unit increase in X**.

In:

$$\begin{aligned} \hat{Y} \\ = 50 + 10X \end{aligned}$$

the slope is:

$$\begin{aligned} \beta_1 \\ = 10 \end{aligned}$$

Therefore, a one-unit increase in X is associated with a **10-unit increase in predicted Y**.

Positive Slope

$$\begin{aligned} \beta_1 \\ > 0 \end{aligned}$$

As X increases, predicted Y increases.

Negative Slope

$$\begin{aligned} \beta_1 \\ < 0 \end{aligned}$$

As X increases, predicted Y decreases.

Zero Slope

$$\begin{aligned} \beta_1 \\ = 0 \end{aligned}$$

There is no linear change in predicted Y as X changes.

Complete Example of Coefficient Interpretation

Suppose a model predicts salary from years of experience:

$$\text{Salary} = 25,000 + 5,000(\text{Experience})$$

Intercept

25,000

When experience is 0 years, the model predicts a salary of ₹25,000.

Slope

5,000

For each additional year of experience, predicted salary increases by approximately **₹5,000**, according to the fitted model.

Quick Revision Table

Topic	Key Point
SLR	One X predicts one Y
Equation	$(Y = \beta_0 + \beta_1 X + \epsilon)$
Intercept	Predicted Y when X = 0
Slope	Change in predicted Y per unit X

Least Squares	Minimizes squared residuals
Logistic Regression	Classification/probability model
MAE	Average absolute error
MSE	Average squared error
RMSE	Square root of MSE
R ²	Proportion of target variation explained
Residual	Actual – Predicted

Important Formulas

Simple Linear Regression

$$\hat{Y} = \beta_0 + \beta_1 X$$

Residual

$$e = Y - \hat{Y}$$

$\wedge$

Least Squares Objective

$$\min \sum (Y - \hat{Y})^2$$

MAE

$$\text{MAE} = \frac{1}{n} \sum |Y - \hat{Y}|$$

MSE

$$\text{MSE} = \frac{1}{n} \sum (Y - \hat{Y})^2$$

RMSE

RMSE=

MSE

-

-

-

-

-

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R²

R

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S

S

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One-Line Memory Trick

SLR → One X, One Y → Fit a straight line → Minimize squared errors → Evaluate with MAE/MSE/RMSE/R².
